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# SciScore Report

Below you will find your SciScore report containing three tables. Your score is calculated based on adherence to scientific rigor criteria (Table 1) and identification of key biological resources (Table 2). Table 3 contains statistical tests and oligonucleotides but is not scored. If SciScore makes any mistakes, please contact us to help us learn and improve.

**Table 1: Rigor Adherence Table**

| |
|---|
| Ethics |
| Consent: All the patients provided written consent for their specimens to be used in the study. |
| IRB: The study was approved by the Ethics Committee of the First Affiliated Hospital of Hainan Medical University |
| Inclusion and Exclusion Criteria |
| not detected. |
| Attrition |
| not detected. |
| Sex as a biological variable |
| not detected. |
| Subject Demographics |
| Age: not detected. |
| Weight: not detected. |
| Randomization |
| not detected. |
| Blinding |
| not detected. |
| Power Analysis |
| not detected. |
| Replication |
| All assays were performed in triplicate. |
| Number: All assays were performed in triplicate. |

| Cell Line Authentication | |
|---|---|
| not detected. | |
| Data Information | |
| Availability: The obtained microarray data were deposited in the Gene Expression Omnibus database ( accession no. GSE101502) | |
| Identifiers: The obtained microarray data were deposited in the Gene Expression Omnibus database ( accession no. GSE101502) | GSE101502 |
| Identifiers: Total RNA was reverse transcribed into first-strand cDNA by incubating at 42degC for 15 min using the RNA first-strand cDNA kit ( cat. no. KR118 TIANGEN Biotech Co . , Ltd. ) and the TaqMan MicroRNA Reverse Transcription kit ( cat. no. 4366596 Thermo Fisher Scientific , Inc . ) . | KR118 |

**Table 2: Key Resources Table**

| Your Sentences | REAGENT or RESOURCE | SOURCE | IDENTIFIER |
|---|---|---|---|
| Antibodies | | | |
| The primary antibodies used were: RAC1 ( cat. no. ab155938 1:200) , GAPDH ( cat. no. ab9485 1:2,000) , MMP9 ( cat. no. ab76003 1:500) , extracellular signal-regulated kinase ( ERK)12 ( cat. no. ab184699 1:500) , pERK12 ( cat. no. ab201015 1:500) , | GAPDH | | |
| | MMP9 | | |
| | pERK12 | | |
| After washing the membranes three times , horseradish peroxidase-conjugated secondary antibodies ( goat anti-Rabbit , cat. no. ab150077 , 1:2,000 and anti-mouse IgG , cat. no. ab150113 , 1:2,000 ) were added , and the membranes were incubated at 25degC for 0.5 h . | anti-Rabbit | | |
| | anti-mouse IgG | | |
| Experimental Models: Cell Lines | | | |
| Cell culture The human colon cancer cell lines , SW620 and 293T , were purchased from the China Center for Type Culture Collection . | SW620 | | Suggestion: RRID:CVCL_0547( link) |
| The 293T cells were seeded at 5x104 cells per well into 24-well plates , co-transfected with 5 ng reporter vectors and 40 pmol miR-142-3p mimics ( UGUAGUGUUUCCUACUUUAUGGA ) or miR-142-3p negative control ( NC ) mimics ( UUUGUACUACACAAAAGUACUG both Guangzhou RiboBio Co . , Ltd. ) using Lipofectamine(r ) 2000 ( Invitrogen Thermo Fisher Scientific , Inc . ) . | 293T | | Suggestion: RRID:CVCL_0063( link) |
| Software and Algorithms | | | |
| The obtained microarray data were deposited in the Gene Expression Omnibus database ( accession no. GSE101502) | Gene Expression Omnibus | | Suggestion: Gene Expression Omnibus (GEO) (RRID:SCR_005012)( link) |
| The target genes of differentially expressed miRNAs were predicted by the PicTar ( 24) , | PicTar | | Suggestion: PicTar (RRID:SCR_003343)( link) |
| The regulatory network of miR-142-3p and its targeted genes were visualized using Cytoscape ( 27) | Cytoscape | | Suggestion: Cytoscape (RRID:SCR_003032)( link) |
| Tripartite overlapping genes were identified , and enriched pathways and cellular functions were identified by Gene Ontology ( GO ) and Kyoto Encyclopedia of Genes and Genomes | KEGG | | |
| | DAVID | | Suggestion: DAVID (RRID:SCR_001881)( link) |

| | | | |
|---|---|---|---|
| ( KEGG ) analyses ( david.ncifcrf . govtools . jsp ) using the Database for Annotation , Visualization , and Integrated Discovery ( DAVID ) v6.8 ( 28) . | | | |
| (Tanon Science and Technology Co., Ltd.) and intensities were quantified using ImageJ v1.8.0 (imagej.nih.govij). | ImageJ | | Suggestion: ImageJ, RRID:SCR_003070( link) |
| Statistical analysis IBM SPSS Statistics for Windows, version 22.0 (IBM Corp.) was used for the statistical analysis. | SPSS | | Suggestion: SPSS, RRID:SCR_002865( link) |

**Other Entities Detected**

| Your Sentences | Recognized Entity |
|---|---|
| Statistical Tests | |
| Students t-test (paired or unpaired depending on the data) was used for the analysis of continuous variables (mean +- standard deviation). | Students t-test |
| One-way analysis of variance with Tukeys post hoc test was used to evaluate the differences between multiple groups. | analysis of variance |
| | Tukeys post hoc test |
| Associations among pathological features (categorical variables) were evaluated using the kh2 and Fishers exact tests. | Fishers exact tests |
| Correlations were assessed using the Pearson correlation coefficient. | Pearson correlation |

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For a full description of scored criteria and tips for improving your score, please see https://www.scicrunch.com/sciscorereport-faq